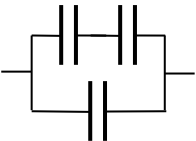
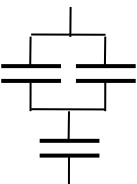


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Substitution into: $C = \frac{\epsilon_0 A}{d}$ (1) Correct answer = 1.24 n[F] (1)	1	1		2	1	
	(b)	(i)	Combining $0.5QV$ and $Q = CV$ (or using $0.5CV^2$) (1) Correct answer = 20 [V] (1)		2		2	2	
		(ii)	2 capacitors in series all in parallel with another  Allow 1 mark for  Subtract a mark if the symbol for capacitor is wrong			2	2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Either opposite charges or attractive force [on plates]		1		1		
		(ii)	Either work done (1) and conservation of energy applied (or implied) (1) OR $\text{energy} = \frac{1}{2} \frac{Q^2}{C}$ (1) and C decreases (1) OR $\frac{Q}{V} = \frac{\epsilon_0 A}{d}$ so $V \propto d$ (1) $E = \frac{1}{2} QV$ hence correct (1) OR energy stored in field (1) field has greater volume (1) OR increased the potential energy (1) Due to attractive force (1)		2		2		
		(iii)	2 or 3 valid equations used e.g. $\frac{1}{2} \frac{Q^2}{C}$ and $C = \frac{\epsilon_0 A}{d}$ (1) Convincing working leading to $\frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$ (if they quote the 2 equations above then simply writing the equation is acceptable) (1)		2		2	2	
		(iv)	Work = force $\times$ distance quoted (1) Hence $F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$ or distance = d and Bethan correct OR Bethan wrong at large separations or when $d = 0$ OR good conclusion well argued (1)			2	2	1	
			Question 4 total	1	8	4	13	8	0